

THE INVENTION OF THE LIGHT BULB

Many inventors helped turn a glowing experiment into light for homes.

On a winter evening in 1879, a glass bulb began to glow at Menlo Park, New Jersey. Thomas Edison and the workers in his laboratory watched the tiny thread of light. They had been chasing that glow for more than a year.

They were not the first people to make electric light. Inventors had experimented with it for decades. In Britain, Joseph Swan was making a practical carbon lamp too, and he demonstrated his lamp in 1879.

At Menlo Park, Edison worked with a team nicknamed the muckers. They tried one material after another for the filament, the thin part that glows. Platinum, rubber and even soot were tested. Most attempts shone briefly, then went dark.

1,000 WAYS

People later told the story this way: Edison had not failed 1,000 times. He had found 1,000 ways that did not work.

A SMALL BULB. A VERY BIG SYSTEM.

Edison's practical light worked because the lamp and the power system were designed together.



EDISON CARBON-FILAMENT LAMP
c. 1880 · Smithsonian (CC0)

The exact number and wording are uncertain. But the real work was just as stubborn: hundreds of experiments and thousands of material tests. Each failure gave the team another clue about what to try next.

Then they tried a small cotton thread. They baked it until it became carbon, sealed it inside a glass bulb and pumped out the air. The filament glowed for at least 13 hours. At last, the light stayed on.

A working bulb was not enough. The team also needed generators, wires, switches, fuses, meters and sockets. They built a whole system so people could use electric light safely.

In 1882, Edison's Pearl Street station began sending power to customers in lower Manhattan. Rooms that had once been lit by candles or gas could glow with electric light.

The change did not happen overnight. Most families kept using candles and gas lamps for many years. But the glowing buildings at Menlo Park had shown that an electric-light system really could work.

So who invented the light bulb? There is no one-person answer. Swan, Edison, their teams and earlier experimenters all helped. The light bulb was not one sudden bright idea. It was a long chain of dim tries, careful notes, teamwork - and one glow that finally lasted.